

15 *Use of the Data Booklet is relevant to this question.*

The reaction of methylpropane with a halogen, X_2 , produces a halogenoalkane.



Which statement about this reaction is correct?

- A** $\text{C}_4\text{H}_9\text{X}$ is formed in one of the propagation reactions.
- B** Small amounts of $(\text{CH}_3)_3\text{C}^\bullet$ are produced in a termination reaction.
- C** The $\text{Br}-\text{Br}$ bond is weaker than the $\text{Cl}-\text{Cl}$ bond and so the reaction with bromine is more exothermic.
- D** The initiation reaction produces a halide ion which is reactive.